

Task Flow-A Team Management Service

Member	Role
Rishanth	Scrum Master, Frontend Developer
Rohith	Product Owner, UI/UX Designer
Veda	Frontend Developer
Suraj	Backend Developer
Abhishek	Testing & Quality Assurance (QA)
Surya	Deployment & DevOps
Akash	Deployment & DevOps

Project:Task Flow

Objective

Develop a web-based platform for team collaboration and task management that enables users to create projects, assign tasks, track progress, communicate in real time, and improve workflow efficiency within teams.

Scrum Roles

Product Owner

- Maintains Product Backlog
- Prioritizes user stories
- Defines acceptance criteria

Scrum Master

- Conducts Daily Scrum
- Facilitates Sprint Planning
- Removes impediments
- Ensures Scrum practices are followed

Development Team

- Designs, develops, tests, and deploys the application
- Delivers potentially shippable increments every sprint

Sprint Plan

Sprint 1 (Weeks 1-2)

Goal: Project Setup & Requirements

Tasks:

- Requirement gathering
- User story creation
- Architecture design
- Database design
- Git repository setup
- Development environment setup

Deliverables:

- Product Backlog
- System Design Document
- Database Schema
- Development Environment

Sprint 2 (Weeks 3-4)

Goal: Authentication & User Management

Tasks:

- User Registration
- Login/Logout
- JWT Authentication
- OTP Verification
- Google Authentication
- Role-based Access Control
- User Profile Management

Deliverables:

- Authentication Module
- User Management Module

Sprint 3 (Weeks 5-6)

Goal: Project & Task Management

Tasks:

- Create Project
- Create and Assign Tasks
- Update Task Status
- Task Priority Management
- Team Management
- Dashboard Development
- Notification System

Deliverables:

- Project Management Module
- Task Management Module
- Dashboard Module

Sprint 4 (Weeks 7-8)

Goal: Real-Time Communication, Deployment & Testing

Tasks:

- Socket.IO Chat Integration
- Real-time Notifications
- Docker Containerization
- CI/CD Setup
- Automated Testing
- Cloud Deployment
- Monitoring and Logging

Deliverables:

- Chat Module
- Production-ready Application
- Automated Deployment Pipeline
- Final Testing Report

Daily Scrum (15 Minutes)

Each member answers:

1. What will I work on today?
2. What will I work on today?
3. Are there any blockers?

Example:

Member	Yesterday	Today	Blocker
Frontend Developer	Login UI	Dashboard UI	None
Backend Developer	Authentication API	Asset API	None
QA Engineer	Login Testing	API Testing	Waiting for API

DevOps Tools

Activity	Tool
Version Control	Git
Repository	GitHub
Project Management	Jira / GitHub Projects
Frontend Development	React
Backend Development	Node.js, Express.js
Database	PostgreSQL
Authentication	JWT, Google OAuth
Real-Time Communication	Socket.IO
API Testing	Postman
CI/CD	GitHub Actions
Containerization	Docker
Monitoring	Docker Logs
Cloud Deployment	Render / Railway

Team_lota

Sprint	Duration	Goal	Rishanth (Scrum Master & Frontend)	Rohith (Product Owner & UI/UX)	Veda (Frontend)	Suraj (Backend)	Abhishek (Testing)	Surya (Deploy ment)	Akash (Deploy ment)								
Sprint 1	2 Weeks	Requirements Analysis & Authentication	Sprint Planning, Git Setup, Frontend Setup	Requirement Analysis, User Stories, Product Backlog, UI Wireframes	Login UI, Forgot Password UI, API Integration	Database Design, JWT Authentication, Admin User Creation, Login API, OTP Verification, Google OAuth, User Management APIs, Team APIs, Invitation APIs, Transfer	Authentication Testing, API Testing, Test Cases	Development Environment Setup	Git Repository Management								
Sprint 2	2 Weeks	User & Team Management	Dashboard UI, Team Management Pages	UI Review, Feature Validation	Employee Management UI, Team Pages	User Management APIs, Team APIs, Invitation APIs, Transfer	Team Module Testing, Integration Testing	Deployment Configuration	Docker Research & Environment Setup								
Sprint 3	2 Weeks	Task Management & Productivity	Task Dashboard, Task UI Integration	UX Improvements	Task Pages, File Upload UI, Progress UI	Task APIs, Subtask APIs, Multi-File Upload, Progress Tracking APIs, Dashboard, Socket.io	Functional Testing, File Upload Testing	Docker Containerization	Docker Compose Configuration								
Sprint 4	2 Weeks	Chat, Notifications & Deployment	Chat UI, Notification UI, Sprint Review	Final Product Review	Chat Pages, Notification Pages, UI Polish	Backend, Notifications API, Reports API, Backend	System Testing, Regression Testing, Final QA	Render Deployment, Docker Deployment	Jenkins CI/CD Pipeline, Deployment Automation								

Team Alpha – Detailed Roles and Responsibilities

Project: Task Flow – Team Collaboration and Task Management Platform

Member 1 – Scrum Master

Primary Responsibility: Ensure the team follows Scrum practices and completes sprint goals on time.

Detailed Responsibilities:

- Conduct Sprint Planning, Daily Scrum, Sprint Review, and Retrospective meetings.
- Monitor sprint progress using dashboards and reports.
- Coordinate among developers, testers, DevOps, and deployment teams.
- Identify blockers and risks and help remove them.
- Track sprint velocity and team productivity.
- Ensure user stories are completed within sprint deadlines.
- Maintain sprint backlog visibility.
- Communicate project status to stakeholders.

Deliverables:

Sprint Plans, Sprint Reports, Burndown Charts, Retrospective Reports, Scrum Dashboard Updates.

Member 2 – Product Owner

Primary Responsibility: Define system requirements and prioritize features based on user needs.

Detailed Responsibilities:

- Gather requirements from users and stakeholders.
- Define project scope, objectives, and success criteria.
- Create and maintain the product backlog.
- Prioritize features such as authentication, project management, task management, chat, and notifications.
- Write user stories and acceptance criteria.
- Review completed features and validate requirements.
- Manage requirement changes throughout development.

Example User Stories:

- User Registration and Login

- Project Creation and Management
- Task Assignment and Tracking
- Team Chat
- Notifications
- Dashboard Analytics

Deliverables:

Product Backlog, User Stories, Acceptance Criteria, Requirement Specifications.

Member 3 – DevOps Engineer

Primary Responsibility: Automate building, testing, and deployment of the application.

Detailed Responsibilities:

- Create and maintain Git repositories.
- Define branching strategies and manage pull requests.
- Configure GitHub Actions workflows.
- Create CI/CD pipelines.
- Build and manage Docker images.
- Configure Docker Compose for local development.
- Automate testing and deployment workflows.
- Manage environment variables and staging environments.
- Maintain build scripts and deployment automation.

Pipeline Flow:

Code Commit → Build → Test → Docker Image → Deploy

Deliverables:

Git Repository Structure, CI/CD Pipeline, Docker Configuration, Build Automation Scripts, Deployment Documentation.

Member 4 – Frontend Developer

Primary Responsibility: Develop the user interface of the Task Flow platform.

Detailed Responsibilities:

- Develop Login and Registration pages.
- Create Project and Task Management interfaces.
- Build Dashboard and Analytics views.

- Implement Team Chat and Notification interfaces.
- Create User Profile and Settings pages.
- Ensure responsive design across devices.
- Integrate frontend with backend APIs.
- Implement form validation and protected routes.
- Improve usability and user experience.

Dashboard Features:

- Total Projects
- Total Tasks
- Completed Tasks
- Pending Tasks
- Team Members

Deliverables:

Web Pages, Responsive Design, API Integration, Frontend Components.

Member 5 – Backend Developer

Primary Responsibility: Develop server-side functionality and business logic.

Detailed Responsibilities:

- Design PostgreSQL database schema.
- Normalize database tables and relationships.
- Develop REST APIs using Node.js and Express.js.
- Implement JWT Authentication.
- Implement OTP Verification.
- Integrate Google OAuth Login.
- Manage sessions and cookies.
- Develop Project and Task Management APIs.
- Implement Socket.IO for real-time chat.
- Create notification services.
- Implement role-based access control.
- Handle validation, logging, and error management.
- Secure APIs and sensitive data.

Database Tables:

Users, Projects, Tasks, Messages, Notifications.

API Modules:

Authentication APIs, Project APIs, Task APIs, Chat APIs, Notification APIs.

Deliverables:

Database Schema, REST APIs, Business Logic Modules, Security Modules.

Member 6 – QA/Test Engineer

Primary Responsibility: Ensure the system works correctly and is free from defects.

Detailed Responsibilities:

- Prepare Test Plans and Test Cases.
- Design Test Scenarios.
- Perform Functional Testing.
- Conduct API Testing using Postman.
- Perform User Acceptance Testing.
- Execute Regression Testing.
- Conduct Cross-browser and Responsive Testing.
- Verify authentication, task management, chat, and notifications.
- Track bugs and verify fixes.
- Prepare quality reports.

Deliverables:

Test Plans, Test Cases, Bug Reports, Test Summary Reports.

Member 7 –Deployment

Primary Responsibility: Deploy and maintain the application infrastructure.

Detailed Responsibilities:

- Configure Docker containers.
- Deploy applications to Render or Railway.
- Manage production releases.
- Maintain deployment documentation.
- Monitor application availability and health.
- Track logs and deployment status.
- Manage backups and recovery procedures.
- Configure environment variables.
- Implement access control and deployment security.
- Maintain rollback and release strategies.

Deliverables:

Production Environment, Deployment Configuration, Monitoring Reports, Backup Strategy, Release Documentation.